

Game Environment

Paradigm

Benchmark Evidence

Taxonomy



L5

Cross-game generalization



L4

Visual agency



L3

Social intelligence



L2

Strategic reasoning



L1

Rule understanding

Probes

L5

first-contact transfer

L4

screen to action

L3

coordination and deception

L2

adaptive planning

L1

legal action



agent

Interaction Paradigm

Observation



text / symbolic



mixed (scaffolded)



raw visual

Action



semantic (API)



native (control)

Evaluation Paradigm

Metric paradigms



outcome (result)



process (trajectory)



adversarial (competitive)

Calibration & robustness



anchors (calibrated)



noise (robust)



held-out (private)



Capability coverage

- multi-level coverage
- errorlementary pressure



Bottleneck diagnosis

- failure localization
- error decomposition



Score meaning

- calibrated reference
- human / AI anchor



Benchmark validity

- robustness checks
- contamination defense

Structure: form + construction



dialogue



match



encounter



arc



world



mixed



embedded



wrapped



adapted



authored



generated

Scope: single → suite → generated/open-ended



single game



game family



curated suite



expandable suite



open-ended tasks